Combined standard deviation

$$\sigma_{12} = \sqrt{\frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_2^2\right)}{n_1 + n_2}}$$

$$d_1 = \overline{x_{12}} - \overline{x_1}$$

$$d_2 = \overline{x_{12}} - \overline{x_2}$$

Example 6.11

From the analysis of monthly wages paid to employees in two service organizations X and Y, the following results were obtained

	Organization X	Organization Y
Number of wage-earners	550	650
Average monthly wages	5000	4500
Variance of the distribution of wages	900	1600

(i) Which organization pays a larger amount as monthly wages?

(ii) Find the combined standard deviation?

Solution:

(i) For finding out which organization X or Y pays larger amount of monthly wages, we have to compare the total wages:

Total wage bill paid monthly by X and Y is

$X: n_1 \times \bar{x}_1 = 550 \times 5000 = ₹\ 27,50,000$

$Y: n_2 \times \bar{x}_2 = 650 \times 4500 = ₹\ 29,25,000$

Organization Y pays a larger amount as monthly wages as compared to organization X.

(ii) For calculating the combined variance, we will first calculate the combined mean

$$\overline{x_{12}} = \frac{n_1\overline{x_1} + n_2\overline{x_2}}{n_1 + n_2}$$

$$= \frac{2750000 + 29250000}{550 + 650}$$

$$= \text{Rs. } 4729.166$$

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